

Time Series Analysis

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Introduction

The cryptocurrency market has experienced significant growth in recent decades, with Bitcoin, in particular, achieving the highest market capitalisation and demonstrating exponential rise. The objective of this report is to undertake a thorough exploration of the monthly historical performance of the Bitcoin price index (in USD) from August 2011 to January 2025. The objective of this study is to analyse the time series data of monthly prices between 2011 and 2025 in order to ascertain the most appropriate ARIMA model.

Methods

Initially, the time series data of the monthly Bitcoin price will be examined for non-stationarity. This will be achieved by employing the autocorrelation function (ACF) and the partial autocorrelation function (PACF) plots, as well as the hypothesis test. The Dickey-Fuller unit-root test, the Phillips-Perron test and the KPSS test will be utilised in this endeavour. Secondly, the series was confirmed to be normal by applying the QQ plot and the Shapiro-Wilk test. In instances where the series exhibits nonstationarity, a transformation and differencing process is implemented to ascertain the series' stationarity prior to model specification.

Following the confirmation of the stationary series, the ACF, PACF plot, EACF table and BIC table are generated for the purpose of identifying the set of all possible ARIMA models in model specification. The objective of the present study is to ascertain the most appropriate ARIMA model by fitting all possible models and calculating parameter estimates and goodness-of-fit metrics.

Results

Overview of Time Series Data

The following descriptive statistical summary is presented for the purpose of offering a more sophisticated understanding of the time series data.

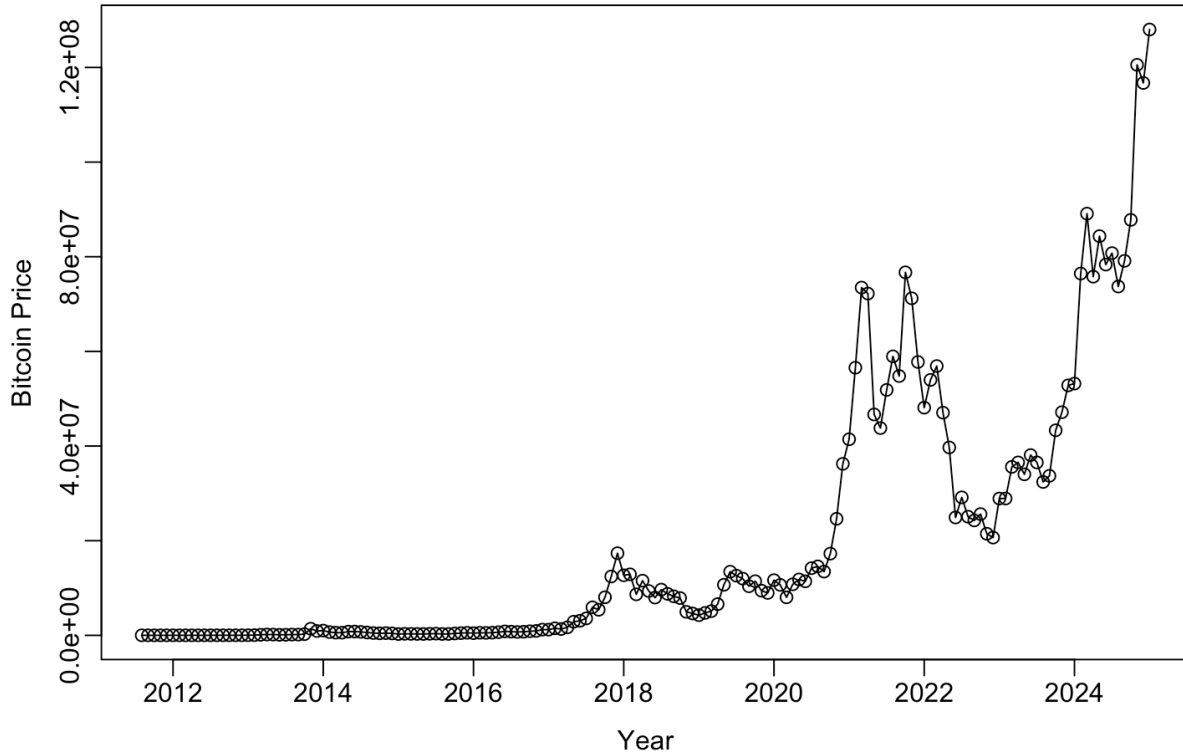
```
bc.ts %>% summary()
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
3988	470981	7930612	19781743	31599541	128015000

Table 1: Descriptive Statistical Summary of Monthly Bitcoin Price

The discrepancy between the maximum and minimum monthly Bitcoin prices is substantial. Furthermore, the mean value exceeding the median value by more than twofold indicates that extreme high variation has had a significant impact.

Figure 1: Time series plot of Monthly Bitcoin Price Series



As demonstrated in Figure 1, the upward quadratic trend is exhibited, thus indicating that the series is nonstationary. The plot demonstrates no clear seasonality, increasing variance over time, and distinct change points around between 2020 and 2022. It is evident that the subject displays fluctuations, and that its behaviour is predominantly autoregressive. This is indicative of Bitcoin's nature as a highly dynamic and market-driven asset.

Measures of Nonstationarity

Figure 2: ACF of Bitcoin Price Change

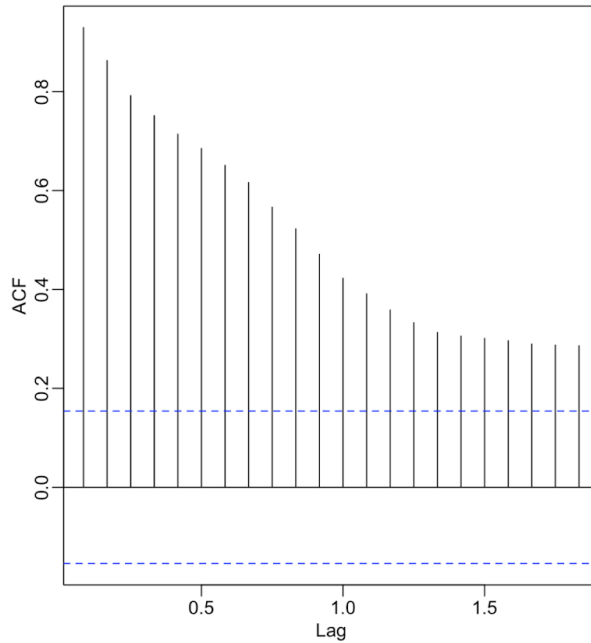
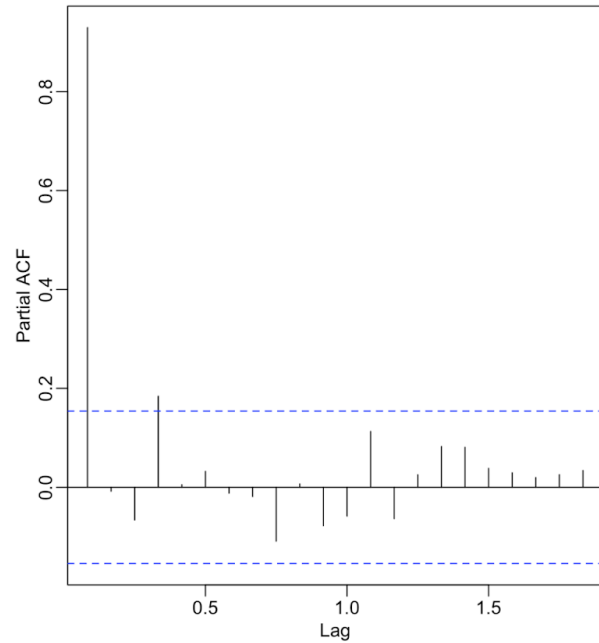


Figure 3: PACF of Bitcoin Price Change



The existence of both ACF and PACF plots indicates that the series under consideration is non-stationary. This is due to the fact that the plot in Figure 2 depicts a slowly decaying pattern, and the high first lag of approximately 1, as can be seen in Figure 3.

In order to ascertain whether the series is non-stationary, hypothesis tests are employed, with an alpha level of 0.05. For the Dickey-Fuller unit-root test and the Phillips-Perron test, the null and alternative hypothesis listed below, while the null and alternative hypotheses switch in the KPSS test.

H_0 : The series is nonstationary but becomes stationary after differencing

H_A : The series is stationary.

Augmented Dickey-Fuller Test

```
data: bc.ts
Dickey-Fuller = -0.27056, Lag order = 5, p-value = 0.99
alternative hypothesis: stationary
```

Table 2: Dickey-Fuller Unit-root Test

```
Phillips-Perron Unit Root Test  
data: bc.ts  
Dickey-Fuller Z(alpha) = -2.4275, Truncation lag parameter = 4, p-value =  
0.9557  
alternative hypothesis: stationary
```

Table 3: Phillips-Perron Test

It is apparent from the results of the ADF and PP tests that the p-values are larger than the alpha level of 0.05, approximately 1 in both cases. This indicates that the series is non-stationary.

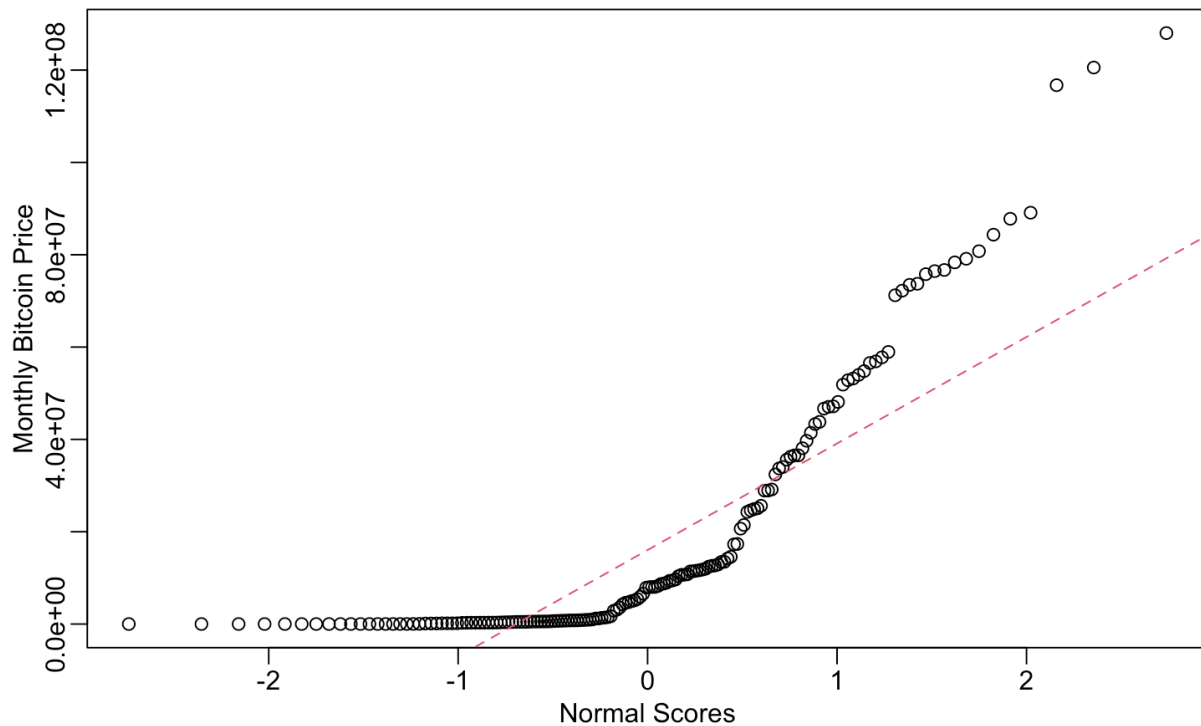
```
KPSS Test for Level Stationarity  
data: bc.ts  
KPSS Level = 2.3395, Truncation lag parameter = 4, p-value = 0.01
```

Table 4: KPSS Test

In contrast, the KPSS test demonstrates a p-value of less than 0.05, thereby confirming the non-stationary nature of this series.

Measures of Normality

Figure 4: QQ plot of Monthly Bitcoin Price



As illustrated in Figure 4, the majority of the data points deviate from the linear trend, indicating that this series does not conform to a normal distribution.

```
Shapiro-Wilk normality test  
  
data:  bc.ts  
W = 0.73793, p-value = 1.102e-15
```

Table 5: Shapiro-Wilk normality Test

The Shapiro-Wilk test yielded a p-value of less than 0.05, thus confirming the previous findings that the series under consideration did not follow a normal distribution.

Transformation and Differencing

In advance of model specification, we have to deal with the nonstationarity by applying transformation and differencing to the raw series.

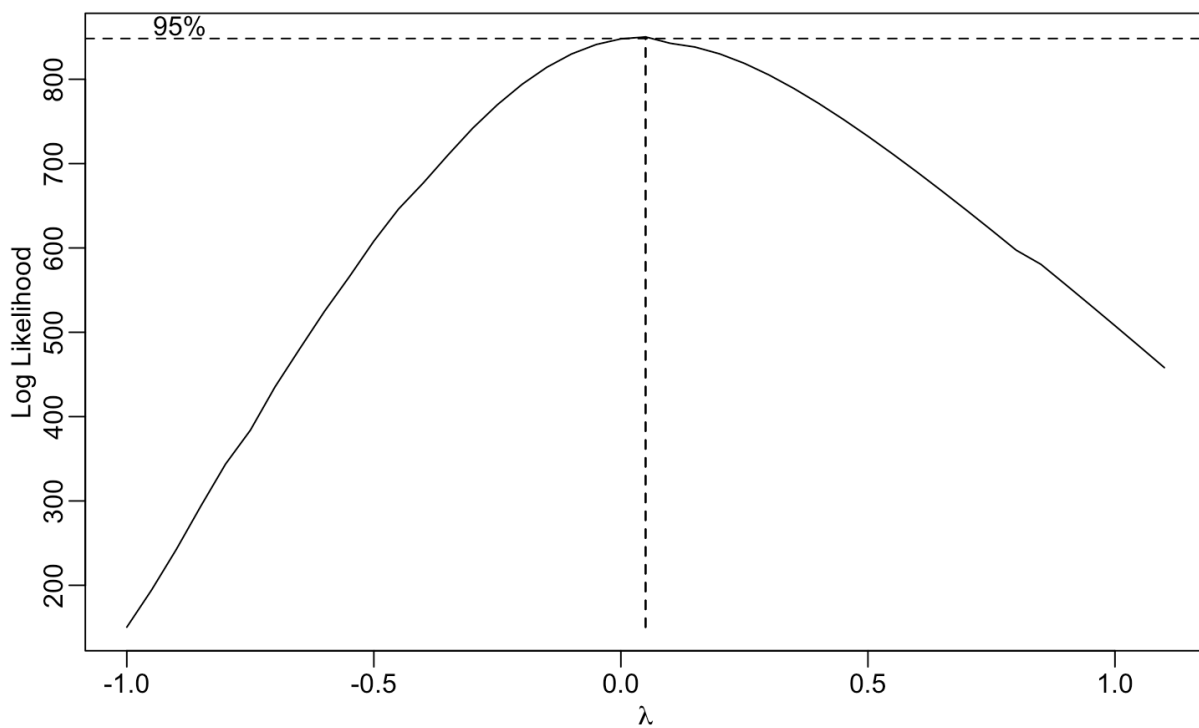


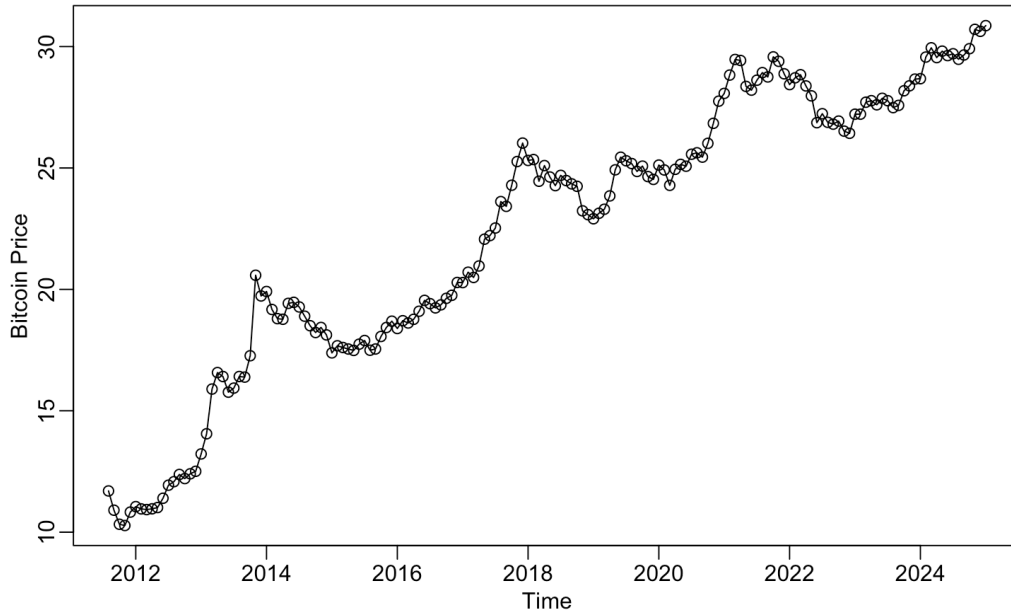
Figure 5: Log Likelihood plot

```
> BC$ci
[1] 0.05 0.05
> lambda <- BC$lambda[which(max(BC$loglike) == BC$loglike)]
> lambda
[1] 0.05
> newBC.ts <- ((bc.ts^lambda)-1)/lambda
```

Table 6: Optimal Lambda

The Box-Cox transformation is applied to the series while obtaining the optimal lambda value of 0.05, which is close to log transformation. Subsequently, we will perform the descriptive analysis of Box-Cox transformed series.

Figure 6: Time Series Plot of Box-Cox Transformed Bitcoin Price



As depicted in Figure 6, an upward quadratic trend is evident, devoid of seasonality and fluctuations in variance. The series displays autoregressive behaviour, with no apparent change point. Nonetheless, the presence of a trend ensures that the Box-Cox transformed series remains nonstationary.

Figure 7: ACF of Box-Cox Transformed Series

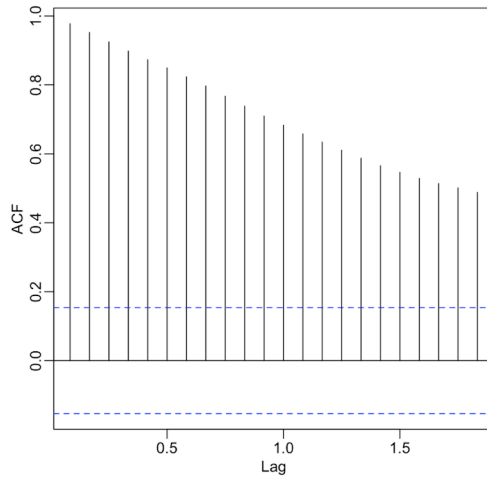
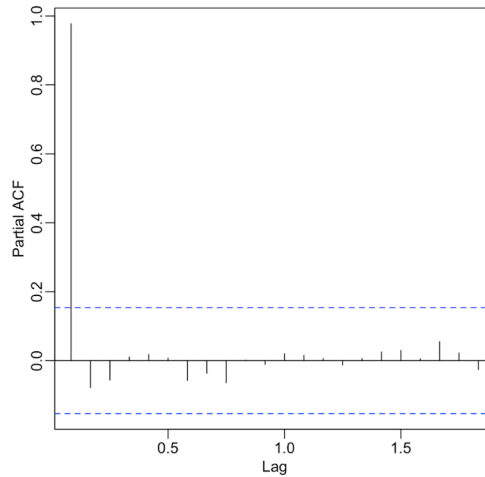


Figure 8: PACF of Box-Cox Transformed Series



The slowly decaying pattern is still ascertained in the ACF plot in Figure 7, as well as the high first lag of approximately 1 in Figure 8. Both are indicating the Box-Cox transformed series is nonstationary.

```

> adf.test(newBC.ts)

        Augmented Dickey-Fuller Test

data:  newBC.ts
Dickey-Fuller = -2.4479, Lag order = 5, p-value = 0.3895
alternative hypothesis: stationary

> pp.test(newBC.ts)

        Phillips-Perron Unit Root Test

data:  newBC.ts
Dickey-Fuller Z(alpha) = -10.418, Truncation lag parameter = 4, p-value =
0.5175
alternative hypothesis: stationary

> kpss.test(newBC.ts)

        KPSS Test for Level Stationarity

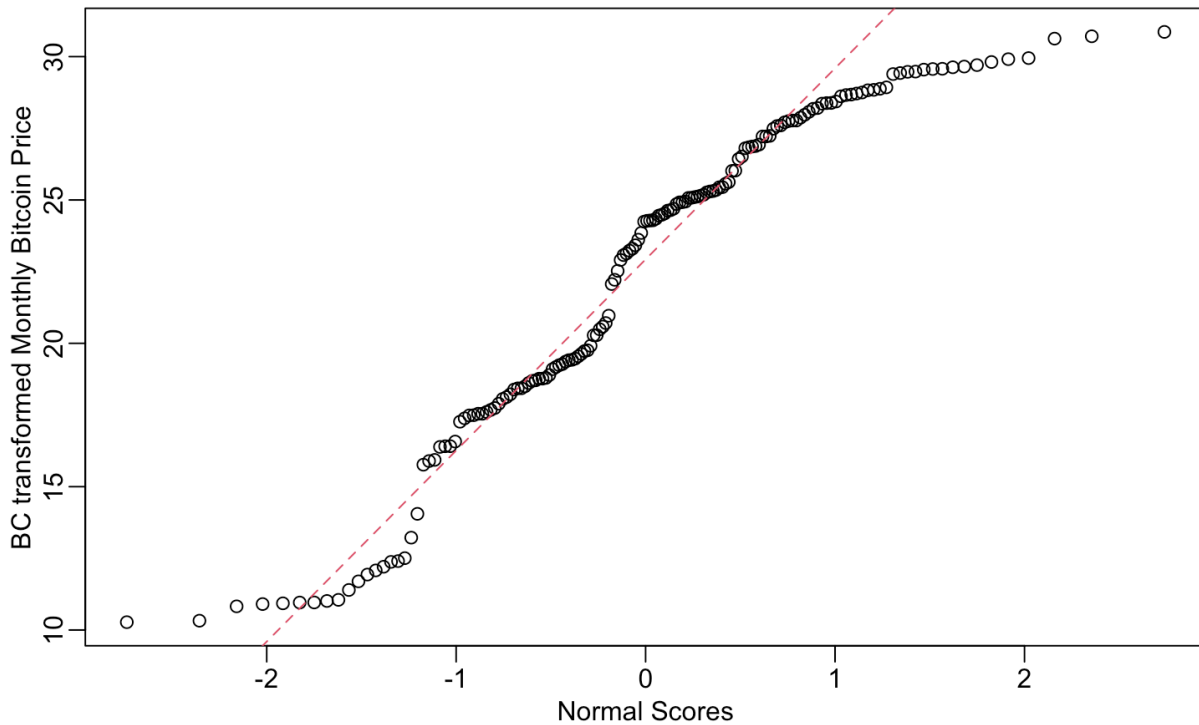
data:  newBC.ts
KPSS Level = 3.0792, Truncation lag parameter = 4, p-value = 0.01

```

Table 7: Hypothesis Test of Nonstationarity for Box-Cox Transformed Series

Similarly, p-values from the ADF test and the PP test, which are larger than 0.05, suggest the transformed series is nonstationary. Furthermore, the outcome of the KPSS test substantiates the hypothesis that the transformed series is non-stationary.

Figure 9: QQ plot of Box-Cox Transformed Bitcoin Price



Shapiro-Wilk normality test

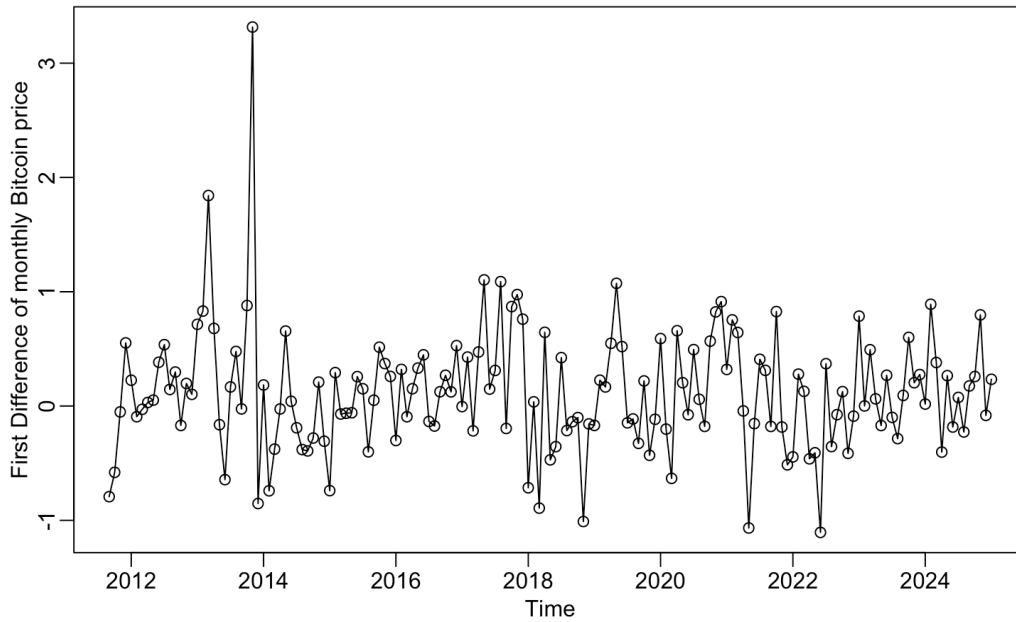
```
data: newBC.ts  
W = 0.93252, p-value = 6.461e-07
```

Table 8: Shapiro-Wilk normality Test for Box-Cox Transformed Series

As demonstrated in Figure 9, the data points align with a linear trend, yet both tails are still off the line. This indicates that the normality of the Box-Cox transformed series has significantly improved. As evidenced by the low p-value from the Shapiro-Wilk test, the transformed series does not conform to a normal distribution, as illustrated in Table 7.

Despite the nonstationarity of the Box-Cox transformed series, the normality has a substantial improvement, and thus the differencing will be conducted with the transformed series.

Figure 10: First difference of Box-Cox Transformed Bitcoin Price Series



Following the first differencing process, the transformed series demonstrated a flat mean level in Figure 10, thereby indicating the absence of any trend.

Figure 11: ACF of First Differencing

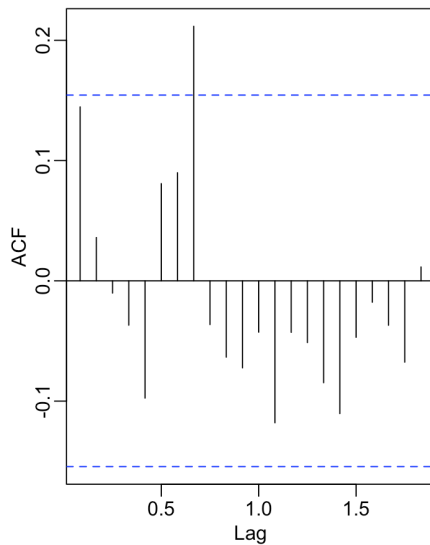
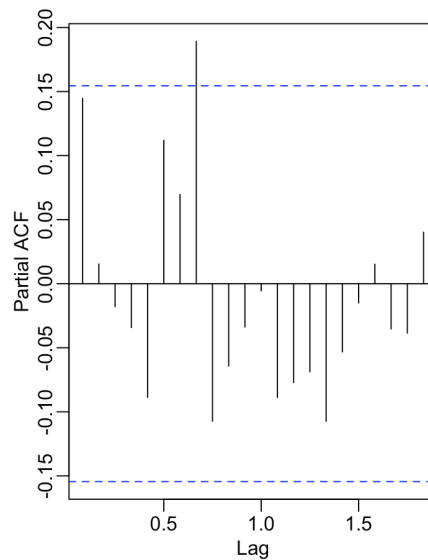


Figure 12: PACF of First Differencing



As shown in Figures 11 and 12, the slowly decaying pattern is eliminated in the ACF plot and the high first lag is no longer evident in the PACF plot, suggesting that the series has become stationary.

```

> adf.test(diff.bc.ts) # p < 0.05 -> stationary

      Augmented Dickey-Fuller Test

data:  diff.bc.ts
Dickey-Fuller = -4.896, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary

Warning message:
In adf.test(diff.bc.ts) : p-value smaller than printed p-value
> pp.test(diff.bc.ts) # p < 0.05 -> stationary

      Phillips-Perron Unit Root Test

data:  diff.bc.ts
Dickey-Fuller Z(alpha) = -135.32, Truncation lag parameter = 4, p-value =
0.01
alternative hypothesis: stationary

Warning message:
In pp.test(diff.bc.ts) : p-value smaller than printed p-value
> kpss.test(diff.bc.ts) #

      KPSS Test for Level Stationarity

data:  diff.bc.ts
KPSS Level = 0.072676, Truncation lag parameter = 4, p-value = 0.1

```

Table 8: Hypothesis Test of Nonstationarity for Box-Cox Transformed Series

In order to confirm the stationarity of the first differenced series, both p-values are less than 0.05 in ADF test and PP test, proving the first differenced series is stationary. Additionally, the result of the KPSS test revealed the stationarity of the series.

Model specification

Following confirmation of the stationarity of the first differenced series, the model specifications for ARIMA models (d=1) are implemented through the application of the ACF, PACF, EACF and BIC tables.

ACF and PACF

Figure 11: ACF of First Differencing

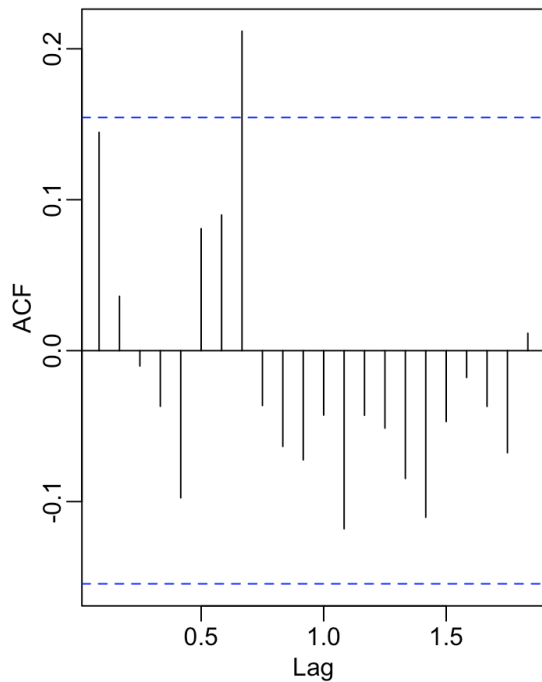
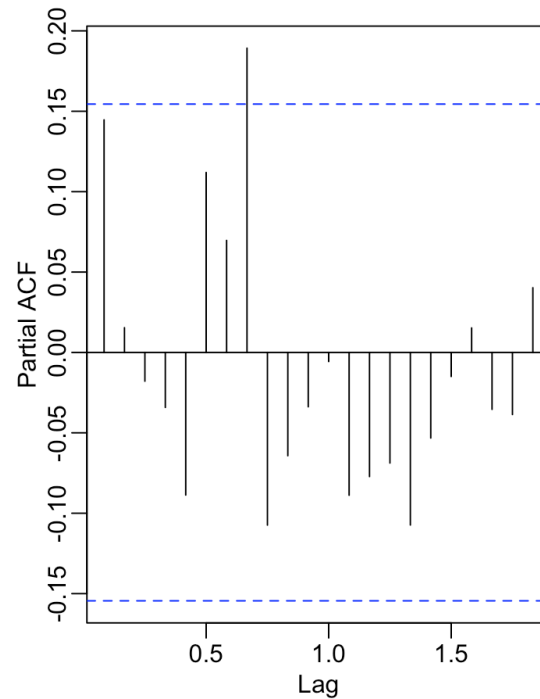


Figure 12: PACF of First Differencing



As demonstrated in Figures 11 and 12, both first lags in the ACF and PACF plots are close to the upper interval, which should be included. Thus, p should be equal to 2 from the PACF plot and q should be equal to 2 from the ACF plot. The potential model is $ARIMA(2,1,2)$, as indicated by the ACF and PACF plots.

EACF

```
> eacf(diff.bc.ts)
```

AR/MA		0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	o	o	o	o	o	o	o	o	x	o	o	o	o	o	o
1	o	o	o	o	o	o	o	o	x	o	o	o	o	o	o
2	x	o	o	o	o	o	o	o	x	o	o	o	o	o	o
3	x	o	o	o	o	o	o	o	x	o	o	o	o	o	o
4	x	o	x	o	o	o	o	o	o	o	o	o	o	o	o
5	x	x	x	o	x	o	o	o	o	o	o	o	o	o	o
6	x	x	x	o	x	o	o	o	o	o	o	o	o	o	o
7	x	x	x	o	x	o	o	o	o	o	o	o	o	o	o

Table 9: EACF of First Differenced Series

The top left 'o' (p, q) from the EACF table should be (0, 0). However, it is imperative to note that the combination of (0, 0) should be excluded and the neighbours should be considered, including (0, 1), (1, 0), and (1, 1). Therefore, the possible models are ARIMA(0, 1, 1), ARIMA(1,1,0) and ARIMA(1, 1, 1).

BIC

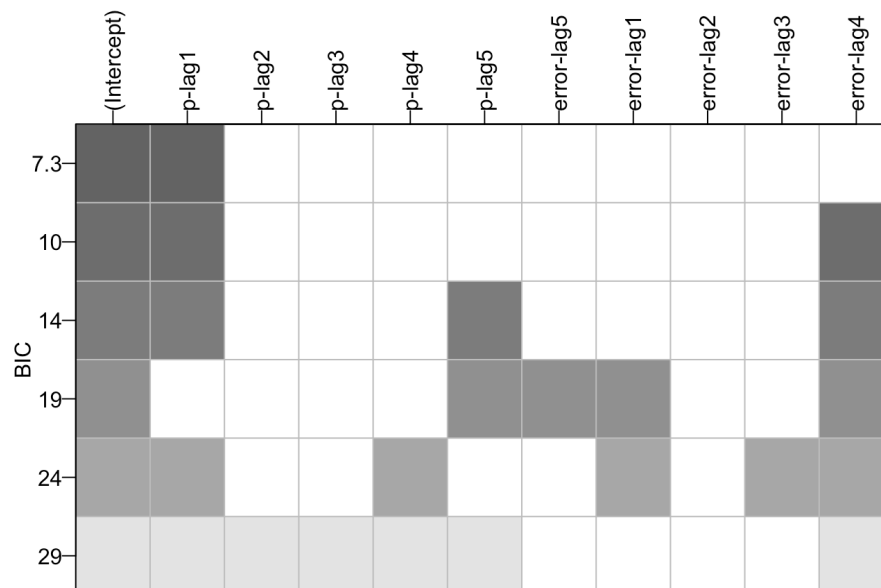


Figure 13: BIC table

Given the modest scale of the potential models from the ACF, PACF and EACF, the maximum values for the BIC table are designated as five.

According to the BIC table, the combination of (1, 0) is derived from the first row and the combination of (1, 4) is obtained from the second row. Moreover, the combination of (5, 4) from the third row should be included, since this is counted as the top three models and the $p = 5$ is still supported. Therefore, the possible models are ARIMA(1, 1, 0), ARIMA(1, 1, 4) and ARIMA(5, 1, 4), as indicated in the BIC table.

Overall, the final set of possible ARIMA model are ARIMA(2, 1, 2), ARIMA(0, 1, 1), ARIMA(1, 1, 0), ARIMA(1, 1, 1), ARIMA(1, 1, 4) and ARIMA(5, 1, 4).

Applications of ARIMA Models with Parameter Estimation

ARIMA(2, 1, 2)

```
> model.212 = Arima(bc.ts, order=c(2,1,2), method='ML')
> coeftest(model.212)
```

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ar1	-0.529398	0.091141	-5.8086	6.301e-09	***
ar2	-0.835877	0.068231	-12.2507	< 2.2e-16	***
ma1	0.662452	0.061559	10.7612	< 2.2e-16	***
ma2	0.935219	0.053817	17.3777	< 2.2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 10: Z Test of ARIMA(2, 1, 2) with Maximum Likelihood Estimation


```

> model.212CSS = Arima(bc.ts, order=c(2,1,2), method='CSS')
> coeftest(model.212CSS)

z test of coefficients:

      Estimate Std. Error  z value  Pr(>|z|)
ar1 -0.545799   0.098730  -5.5282 3.236e-08 ***
ar2 -0.848059   0.067947 -12.4812 < 2.2e-16 ***
ma1  0.676515   0.061934  10.9231 < 2.2e-16 ***
ma2  0.952984   0.053596  17.7810 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Table 11: Z Test of ARIMA(2, 1, 2) with Least Squares Estimation

As demonstrated in Tables 10 and 11, all the p-values of the estimated coefficients are less than 0.05. This finding supports the hypothesis that all the estimated coefficients are statistically significant. This finding suggests that the ARIMA(2,1,2) model exhibited a high degree of suitability for both Maximum Likelihood estimation and Least Squares estimation.

ARIMA(0, 1, 1)

```
> model.011 = Arima(bc.ts, order=c(0,1,1), method='ML')
> coeftest(model.011)

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ma1 0.126071    0.075546  1.6688  0.09516 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
> model.011CSS = Arima(bc.ts, order=c(0,1,1), method='CSS')
> coeftest(model.011CSS)

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ma1 0.126806    0.075739  1.6742  0.09408 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 12: Z Test of ARIMA(0, 1, 1) with Maximum Likelihood Estimation and Least Squares Estimation

With regard to the ARIMA(0, 1, 1) model, p-values of the estimated coefficients are slightly larger than the alpha level of 0.05, irrespective of the Maximum Likelihood estimation and Least Squares estimation methods employed in Table 12. This outcome suggests that the estimated coefficients are not statistically significant in the ARIMA(0, 1, 1) model.

ARIMA(1, 1, 0)

```
> model.110 = Arima(bc.ts,order=c(1,1,0), method='ML')
> coeftest(model.110)

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 0.131888   0.078757  1.6746  0.09401 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
> model.110CSS = Arima(bc.ts,order=c(1,1,0), method='CSS')
> coeftest(model.110CSS)

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 0.132726   0.079026  1.6795  0.09305 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 12: Z Test of ARIMA(1, 1, 0) with Maximum Likelihood Estimation and Least Squares Estimation

As with the ARIMA(0, 1, 1) model, the p-values of the estimated coefficients in the ARIMA(1, 1, 0) model are marginally greater than the alpha level of 0.05, irrespective of the Maximum Likelihood estimation and Least Squares estimation methods employed in Table 12. This finding also suggests that the estimated coefficients are not statistically significant in the ARIMA(1, 1, 0) model.

ARIMA(1, 1, 1)

```
> model.111 = Arima(bc.ts,order=c(1,1,1), method='ML')
> coeftest(model.111)

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.1393832  0.4591484  0.3036  0.7615
ma1 -0.0062261  0.4607667 -0.0135  0.9892

>
> model.111CSS = Arima(bc.ts,order=c(1,1,1), method='CSS')
> coeftest(model.111CSS)

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.1410742  0.4638558  0.3041  0.7610
ma1 -0.0094778  0.4659000 -0.0203  0.9838
```

Table 13: Z Test of ARIMA(1, 1, 1) with Maximum Likelihood Estimation and Least Squares Estimation

According to the Table 13, all p-values of the estimated coefficients in the ARIMA(1, 1, 1) model are larger than the alpha level of 0.05, indicating all estimated coefficients are not statistically significant in the ARIMA(1, 1, 1) model.

ARIMA(1, 1, 4)

```
> model.114 = Arima(bc.ts,order=c(1,1,4), method='ML')
> coeftest(model.114)
```

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ar1	-0.714550	0.184622	-3.8703	0.0001087	***
ma1	0.896413	0.184164	4.8675	1.13e-06	***
ma2	0.147209	0.113101	1.3016	0.1930639	
ma3	-0.052147	0.118222	-0.4411	0.6591424	
ma4	0.127529	0.105913	1.2041	0.2285527	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
>
> model.114CSS = Arima(bc.ts,order=c(1,1,4), method='CSS')
> coeftest(model.114CSS)
```

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ar1	-0.720819	0.190688	-3.7801	0.0001568	***
ma1	0.903800	0.188593	4.7923	1.649e-06	***
ma2	0.149802	0.114327	1.3103	0.1900960	
ma3	-0.053474	0.120142	-0.4451	0.6562544	
ma4	0.130248	0.108991	1.1950	0.2320716	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 14: Z Test of ARIMA(1, 1, 4) with Maximum Likelihood Estimation and Least Squares Estimation

As shown in Tables 14, only the estimated coefficients of AR1 and MA1 are statistically significant, that p-values are less than 0.05. This finding indicates that the ARIMA(1, 1, 4) models are not effective in fully capturing the dynamics of the series.

ARIMA(5, 1, 4)

```
> model.514 = Arima(bc.ts,order=c(5,1,4), method='ML')
> coeftest(model.514)

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.183839   0.565365 -0.3252  0.745054
ar2 -0.595264   0.132239 -4.5014 6.750e-06 ***
ar3 -0.628636   0.234090 -2.6855  0.007243 **
ar4  0.098937   0.318938  0.3102  0.756402
ar5 -0.166170   0.190713 -0.8713  0.383584
ma1  0.323349   0.582551  0.5551  0.578855
ma2  0.748503   0.154369  4.8488 1.242e-06 ***
ma3  0.746983   0.363421  2.0554  0.039839 *
ma4  0.203598   0.434018  0.4691  0.638998
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
> model.514CSS = Arima(bc.ts,order=c(5,1,4), method='CSS')
> coeftest(model.514CSS)

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.255843   1.907246 -0.1341  0.893290
ar2 -0.608200   0.226384 -2.6866  0.007219 **
ar3 -0.679935   0.706457 -0.9625  0.335820
ar4  0.055018   1.103095  0.0499  0.960221
ar5 -0.151778   0.610000 -0.2488  0.803503
ma1  0.397511   1.966875  0.2021  0.839836
ma2  0.763502   0.456662  1.6719  0.094540 .
ma3  0.798488   1.203984  0.6632  0.507199
ma4  0.258316   1.497946  0.1724  0.863086
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 15: Z Test of ARIMA(5, 1, 4) with Maximum Likelihood Estimation and Least Squares Estimation
In the ARIMA(5, 1, 4) model, the p-values of the estimated coefficients are predominantly greater than 0.05, indicating that the majority of the estimated coefficients are not statistically

significant. This outcome indicates that the ARIMA(5, 1, 4) models are not effective in fully capturing the volatility of the series.

Goodness-of-fit Metrics

For the purpose of obtaining the most suitable ARIMA models from among all possible fitted models. To this end, performance metrics are applied below.

AIC and BIC

	df	AIC		df	BIC
model.514	10	5474.550	model.110	2	5485.678
model.212	5	5477.276	model.011	2	5485.783
model.114	6	5478.478	model.111	3	5490.760
model.110	2	5479.516	model.212	5	5492.683
model.011	2	5479.620	model.114	6	5496.967
model.111	3	5481.515	model.514	10	5505.364

Table 16: AIC and BIC of Fitted Models

A comparison of Table 16 reveals that the minimum AIC is the ARIMA(5, 1, 4) model, while the minimum BIC is the ARIMA(1, 1, 0) model.

Other Error Measures

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
ARIMA(2,1,2)	718956.8	5691449	2822234	2.497	18.725	0.217	0.003
ARIMA(0,1,1)	710175.5	5855822	2945637	2.550	17.415	0.227	-0.008
ARIMA(1,1,0)	695133.5	5853895	2950675	2.522	17.443	0.227	-0.012
ARIMA(1,1,1)	693599.2	5853884	2951392	2.518	17.446	0.227	-0.014
ARIMA(1,1,4)	654373.5	5684633	2987766	2.448	19.358	0.230	-0.015
ARIMA(5,1,4)	660717.9	5392454	2829687	2.578	19.371	0.218	-0.014

Table 17: Error Measures

With regards to the error measures of Table 17, the minimum RMSE is the ARIMA(5, 1, 4) model and the smallest MAPE is ARIMA(0, 1, 1), while both minimum RMSE and MASE is the ARIMA(2, 1, 2).

Despite the finding that the ARIMA(5, 1, 4) model exhibited the smallest AIC and RMSE, it is a feasible indication of overfitting given its relatively larger size. Furthermore, as demonstrated in Table 15, the ARIMA(5, 1, 4) model is not statistically significant.

Conversely, the ARIMA(2, 1, 2) model offers both minimum RMSE and MASE. Despite the relatively elevated BIC value of the ARIMA(2, 1, 2) model in comparison to other models, the AIC value is the second lowest. This finding suggests that the performance of the ARIMA(2, 1, 2) model is relatively good when considered against all fitted models. As demonstrated in Tables 10 and 11, it is evident that the estimated coefficients of the ARIMA(2, 1, 2) model, when evaluated through Maximum Likelihood and Least Squares estimation methods, are statistically significant. This finding suggests that the ARIMA(2,1,2) model is the most appropriate ARIMA model.

Overparametrised Models

Subsequently, the overparametrised models are ARIMA(3,1,2) and ARIMA(2,1,3), with the objective of identifying whether other potential models may perform better than the ARIMA(2,1,2) model.

ARIMA(3, 1, 2)

```
> coeftest(model.312)

z test of coefficients:

      Estimate Std. Error  z value  Pr(>|z|)
ar1 -1.381851   0.081640 -16.9262 < 2.2e-16 ***
ar2 -0.705607   0.124508  -5.6672 1.452e-08 ***
ar3  0.019240   0.081513   0.2360  0.8134
ma1  1.616030   0.037926  42.6102 < 2.2e-16 ***
ma2  0.999993   0.043124  23.1889 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
> model.312CSS = Arima(bc.ts, order=c(3,1,2), method='CSS')
> coeftest(model.312CSS)

z test of coefficients:

      Estimate Std. Error  z value  Pr(>|z|)
ar1 -1.6411345   0.0792131 -20.7180 < 2.2e-16 ***
ar2 -0.5973248   0.1451102  -4.1164 3.849e-05 ***
ar3  0.1336300   0.0820282   1.6291   0.1033
ma1  1.8957234   0.0092146 205.7309 < 2.2e-16 ***
ma2  1.0275865   0.0110302  93.1613 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
> model.312CSSML = Arima(bc.ts, order=c(3,1,2), method='CSS-ML')
> coeftest(model.312CSSML)

z test of coefficients:

      Estimate Std. Error  z value  Pr(>|z|)
ar1 -1.641968   0.080472 -20.4043 < 2.2e-16 ***
ar2 -0.578247   0.146721  -3.9411 8.11e-05 ***
ar3  0.148599   0.089284   1.6643  0.09605 .
ma1  1.881087   0.036477  51.5698 < 2.2e-16 ***
ma2  0.999998   0.035828  27.9110 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 18: Z Test of ARIMA(3, 1, 2) with Maximum Likelihood Estimation, Least Squares Estimation and Combined Estimation

According to Table 18, the performance of the ARIMA(3, 1, 2) model with Combined Estimation has improved by comparing to the results of Maximum Likelihood Estimation and Least Squares Estimation. However, the AR3 is still not statistically significant, which means the performance of the ARIMA(3, 1, 2) model is limited.

ARIMA(2, 1, 3)

```
> model.213 = Arima(bc.ts, order=c(2,1,3), method='ML')
> coeftest(model.213)

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -1.3294946  0.4798830 -2.7705 0.005598 **
ar2 -0.4861121  0.3414513 -1.4237 0.154544
ma1  1.5062441  0.4804563  3.1350 0.001718 **
ma2  0.7347833  0.5013443  1.4656 0.142750
ma3  0.0057355  0.2061036  0.0278 0.977799
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
> model.213CSS = Arima(bc.ts, order=c(2,1,3), method='CSS')
> coeftest(model.213CSS)

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -1.293135  0.399110 -3.2400 0.0011951 **
ar2 -0.450365  0.310025 -1.4527 0.1463148
ma1  1.470780  0.400241  3.6747 0.0002381 ***
ma2  0.689744  0.441206  1.5633 0.1179780
ma3 -0.012099  0.177933 -0.0680 0.9457893
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 19: Z Test of ARIMA(2, 1, 3) with Maximum Likelihood Estimation and Least Squares Estimation

As shown in Tables 19, most of the estimated coefficients are not statistically significant, that p-values are less than 0.05. This result suggests that the ARIMA(2, 1, 3) models are not effective in fully capturing the dynamics of the series.

AIC and BIC

	df	AIC		df	BIC
model.312	6	5473.373	model.312	6	5491.861
model.212	5	5477.276	model.212	5	5492.683
model.213	6	5478.367	model.213	6	5496.855

Table 20: AIC and BIC of Overparametrised Models

A comparison of Table 20 reveals that the minimum AIC and BIC is the ARIMA(3, 1, 2) model, followed by the ARIMA(2, 1, 2) model.

Other Error Measures

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
ARIMA(2,1,2)	718956.8	5691449	2822234	2.497	18.725	0.217	0.003
ARIMA(3,1,2)	685288.7	5514151	3021679	2.308	24.125	0.232	-0.013
ARIMA(2,1,3)	685470.7	5683299	2975986	2.551	19.061	0.229	-0.013

Table 21: Error Measures of Overparametrised Models

With regards to the error measures of Table 2, the minimum RMSE is the ARIMA(3, 1, 2) model, while the minimum MAE, RMSE and MASE is the ARIMA(2, 1, 2).

In order to ascertain the most appropriate model between the ARIMA(3, 1, 2) and the ARIMA(2, 1, 2) model, it was determined that the ARIMA(3, 1, 2) model obtained superior results in terms of AIC and BIC values. Nevertheless, the ARIMA(2, 1, 2) model demonstrates superior performance in terms of overall error measures, exhibiting the lowest MAE, RMSE and MASE values. In particular, it is evident that all the p-values of the estimated coefficients are less than the alpha level of 0.05, with maximum likelihood estimation and least squares estimation employed in Tables 10 and 11, respectively. This finding indicates that the ARIMA (2, 1, 2) model possesses a superior capacity to capture the variability of the series.

Conclusion

The ARIMA(2, 1, 2) model is identified as the most appropriate model for the monthly Bitcoin price series, as evidenced by a comparison of different parameter estimations and goodness-of-fit metrics. Although certain metrics demonstrate inconsistencies in the

performance of disparate ARIMA models. The present study has demonstrated that the ARIMA (2, 1, 2) model continues to demonstrate reliable overall performance, as evidenced by the evaluation results.

Appendices

```
sort.score <- function(x, score = c("bic", "aic")){
  if (score == "aic"){
    x[with(x, order(AIC)),]
  } else if (score == "bic") {
    x[with(x, order(BIC)),]
  } else {
    warning('score = "x" only accepts valid arguments ("aic","bic")')
  }
}

# Time series plot of raw series
plot(bc.ts,type='o',ylab='Bitcoin Price', xlab='Year', main='Figure 1: Time series plot of Monthly
Bitcoin Price Series')
# Upwards quadratic trend exists,
# No seasonality, change point
# slightly changing variance, but mostly autoregressive behavior

par(mfrow=c(1,2),
    mar = c(3, 3, 4, 1), # Adjust margin of plots
    mgp = c(1.5, 0.5, 0)) # Adjust xlab/ylab position
acf(bc.ts, main = "Figure 2: ACF of Bitcoin Price Change") # slowly decaying pattern ->non
stationary
pacf(bc.ts, main = "Figure 3: PACF of Bitcoin Price Change") # high first lag ->non stationary
par(mfrow=c(1,1))

# To confirm the stationarity
adf.test(bc.ts) # p-value > 0.05 ->non stationary
pp.test(bc.ts) # p-value > 0.05 ->non stationary
kpss.test(bc.ts) # p-value < 0.05 ->non stationary

# Check normality
qqnorm(bc.ts, ylab="Monthly Bitcoin Price", xlab="Normal Scores", main = "Figure 4: QQ plot of
Monthly Bitcoin Price") #Check normality
qqline(bc.ts, col = 2, lwd = 1, lty = 2) # data is not normal distribution
```

```

shapiro.test(bc.ts) # data is not normal distribution

# Box-Cox transformation
BC <- BoxCox.ar(bc.ts, lambda = seq(-1, 1.125, 0.05))
BC$ci
lambda <- BC$lambda[which(max(BC$loglike) == BC$loglike)]
lambda
newBC.ts <- ((bc.ts^lambda)-1)/lambda

# optimal lambda is 0.05
# The optimal lambda value is 0.05 which is close to log transformation

# Time series plot of BoxCox transformed Bitcoin price
plot(newBC.ts,type='o', ylab ="Bitcoin Price", main="Figure 6: Time Series Plot of Box-Cox
Transformed Bitcoin Price")

par(mfrow=c(1,2),
    mar = c(3, 3, 4, 1), # Adjust margin of plots
    mgp = c(1.5, 0.5, 0))
acf(newBC.ts, main = "Figure 7: ACF of Box-Cox Transformed Series")
pacf(newBC.ts, main = "Figure 8: PACF of Box-Cox Transformed Series")
par(mfrow=c(1,1))
# Upward quadratic trend exists
adf.test(newBC.ts) # p-value > 0.05 ->non stationary
pp.test(newBC.ts) # p-value > 0.05 ->non stationary
kpss.test(newBC.ts) # p-value < 0.05 ->non stationary

# Check normality again
qqnorm(newBC.ts, main = "Figure 9: QQ plot of Box-Cox Transformed Bitcoin Price", ylab="BC
transformed Monthly Bitcoin Price", xlab="Normal Scores")
qqline(newBC.ts,col = 2, lwd = 1, lty = 2) # slightly improvement of the normality, but both tails
are still off
shapiro.test(newBC.ts) # p<0.05 -> data is not normal distribution

# 1st differencing to BoxCox transformed series, because of the improvement of normality
diff.bc.ts = diff(newBC.ts, differences = 1)
plot(diff.bc.ts, type='o',ylab='First Difference of monthly Bitcoin price',
    main ="Figure 10: First difference of Box-Cox Transformed Bitcoin Price Series") # flat mean
level -> stop differencing

par(mfrow=c(1,2),
    mar = c(3, 3, 4, 1), # Adjust margin of plots

```

```

    mgp = c(1.5, 0.5, 0))
acf(diff.bc.ts, main = "Figure 11: ACF of First Differencing")
pacf(diff.bc.ts, main = "Figure 12: PACF of First Differencing")
par(mfrow=c(1,1))

# To confirm the stationarity of the first differenced series
adf.test(diff.bc.ts) # p < 0.05 -> stationary
pp.test(diff.bc.ts) # p < 0.05 -> stationary
kpss.test(diff.bc.ts) # p > 0.05 -> stationary

# Model specification with the first differenced series (d = 1)

# ACF & PACF
par(mfrow=c(1,2))
acf(diff.bc.ts, main = "Figure 9: ACF of First Differencing")
pacf(diff.bc.ts, main = "Figure 10: PACF of First Differencing")
par(mfrow=c(1,1))

# From ACF & PACF, p = 2 & q = 2
# possible models: {ARIMA(2,1,2)}

# EACF
eacf(diff.bc.ts)
# From EACF, (0,1), (1,0), (1,1)
# possible models: {ARIMA(0,1,1), ARIMA(1,1,0), ARIMA(1,1,1)}

#Fig 13: BIC table - from ACF, PACF & EACF, max p & q = 2, so set the limit to 5
res = armasubsets(y=diff.bc.ts, nar=5, nma=5,
                  y.name='p', ar.method='ols')
plot(res)
# From BIC table, (1,0) from first row, (1,4) from second row, (5,4) from third row
# possible models: {ARIMA(1,1,0), ARIMA(1,1,4), ARIMA(5,1,4)}

# Hence, Final set of possible model are:
#{ARIMA(2,1,2), ARIMA(0,1,1), ARIMA(1,1,0), ARIMA(1,1,1), ARIMA(1,1,4), ARIMA(5,1,4)}

# ARIMA(2,1,2) ***
model.212 = Arima(bc.ts, order=c(2,1,2), method='ML')
coeftest(model.212)

model.212CSS = Arima(bc.ts, order=c(2,1,2), method='CSS')
coeftest(model.212CSS)

```

```

# ARIMA(0,1,1)
model.011 = Arima(bc.ts, order=c(0,1,1), method='ML')
coeftest(model.011)

model.011CSS = Arima(bc.ts, order=c(0,1,1), method='CSS')
coeftest(model.011CSS)

# Goodness-of-fit metrics
# Check AIC and BIC values of the models to decide the best one within the subset of possible
models.
sort.score(AIC(model.212, model.011, model.110, model.111, model.114, model.514), score =
"aic")
sort.score(BIC(model.212, model.011, model.110, model.111, model.114, model.514), score =
"bic")

# Hence, Final set of possible model are:
#{ARIMA(0,1,1), ARIMA(1,1,0), ARIMA(1,1,1), ARIMA(1,1,4), ARIMA(5,1,4)}

Smodel_212_css <- accuracy(model.212)[1:7]
Smodel_011_css <- accuracy(model.011)[1:7]
Smodel_110_css <- accuracy(model.110)[1:7]
Smodel_111_css <- accuracy(model.111)[1:7]
Smodel_114_css <- accuracy(model.114)[1:7]
Smodel_514_css <- accuracy(model.514)[1:7]
df.Smodels <- data.frame(
  rbind(Smodel_212_css, Smodel_011_css, Smodel_110_css, Smodel_111_css,
Smodel_114_css, Smodel_514_css)
)
colnames(df.Smodels) <- c("ME", "RMSE", "MAE", "MPE", "MAPE",
"MASE", "ACF1")
rownames(df.Smodels) <- c("ARIMA(2,1,2)", "ARIMA(0,1,1)", "ARIMA(1,1,0)", "ARIMA(1,1,1)",
"ARIMA(1,1,4)", "ARIMA(5,1,4)")
round(df.Smodels, digits = 3)

#           ME  RMSE  MAE  MPE  MAPE  MASE  ACF1
#ARIMA(2,1,2) 718956.8 5691449 2822234< 2.497 18.725 0.217< 0.003
#ARIMA(0,1,1) 710175.5 5855822 2945637 2.550 17.415< 0.227 -0.008
#ARIMA(1,1,0) 695133.5 5853895 2950675 2.522 17.443 0.227 -0.012
#ARIMA(1,1,1) 693599.2 5853884 2951392 2.518 17.446 0.227 -0.014
#ARIMA(1,1,4) 654373.5 5684633< 2987766 2.448 19.358 0.230 -0.015
#ARIMA(5,1,4) 660717.9 5392454 2829687 2.578 19.371 0.218 -0.014

```

```

# The best model is ARIMA(2,1,2). So, the overparametrised models
# are ARIMA(3,1,2) and ARIMA(2,1,3)

# ARIMA(3,1,2)
model.312 = Arima(bc.ts, order=c(3,1,2), method='ML')
coeftest(model.312)

model.312CSS = Arima(bc.ts, order=c(3,1,2), method='CSS')
coeftest(model.312CSS)

model.312CSSML = Arima(bc.ts, order=c(3,1,2), method='CSS-ML')
coeftest(model.312CSSML)

# Goodness-of-fit metrics

sort.score(AIC(model.212, model.312, model.213), score = "aic")
sort.score(BIC(model.212, model.312, model.213), score = "bic")

Smodel_212_css <- accuracy(model.212)[1:7]
Smodel_312_css <- accuracy(model.312)[1:7]
Smodel_213_css <- accuracy(model.213)[1:7]
df.Smodels <- data.frame(
  rbind(Smodel_212_css, Smodel_312_css, Smodel_213_css)
)
colnames(df.Smodels) <- c("ME", "RMSE", "MAE", "MPE", "MAPE",
  "MASE", "ACF1")
rownames(df.Smodels) <- c("ARIMA(2,1,2)", "ARIMA(3,1,2)", "ARIMA(2,1,3)")
round(df.Smodels, digits = 3)

```

Reference

Dr Haydar Demirhan, Time Series Analysis Course Note (2025), Module 4 - 6, course note, accessed 25 April 2025.